

Explicit BCH short Khinchin sets at every even exponent

Partial-solution packet for arXiv:1301.2382

11 August 2026

Candidate substantial partial result. For every even $p \geq 2$ there is a deterministic set $V \subset \{-1, 1\}^n$ of optimal cardinality $O_p(n^{p/2})$ whose p th moments of all linear forms agree *exactly* with those on the full cube. The same construction gives deterministic polynomial-size short Khinchin sets for every real $p \geq 1$.

1 The problem

Rudelson's survey asks, for fixed $p \geq 1$, for a small deterministic set $V \subset \{-1, 1\}^n$ and constants $\alpha_p, \beta_p > 0$ such that

$$\alpha_p \|a\|_2 \leq \left(\frac{1}{|V|} \sum_{x \in V} |\langle a, x \rangle|^p \right)^{1/p} \leq \beta_p \|a\|_2 \quad (a \in \mathbb{R}^n). \quad (1)$$

The survey says that deterministic constructions of reasonably small cardinality are unknown. Its probabilistic construction uses $N \geq n^{p/2}$ for $p > 2$, and it notes that this order is optimal.

2 Statement of the result

Let B_s denote any valid upper constant in the classical Khinchin inequality at exponent s .

Theorem 1. *For every $n \geq 1$ the following deterministic constructions exist.*

1. *If $p = 2m$ is even, there is $V_{n,m} \subset \{-1, 1\}^n$ with*

$$|V_{n,m}| \leq [2(n+1)]^m$$

such that, for every $a \in \mathbb{R}^n$,

$$\frac{1}{|V_{n,m}|} \sum_{x \in V_{n,m}} |\langle a, x \rangle|^{2m} = 2^{-n} \sum_{x \in \{-1, 1\}^n} |\langle a, x \rangle|^{2m}. \quad (2)$$

Consequently (1) holds with the classical constants and cardinality $O_p(n^{p/2})$.

2. *If $p \geq 2$ is arbitrary and $2m$ is the least even integer at least p , the same $V_{n,m}$ satisfies*

$$\|a\|_2 \leq \left(\frac{1}{|V_{n,m}|} \sum_{x \in V_{n,m}} |\langle a, x \rangle|^p \right)^{1/p} \leq B_{2m} \|a\|_2.$$

3. *If $1 \leq p \leq 2$, a deterministic set of cardinality at most $[2(n+1)]^2$ satisfies*

$$\frac{1}{\sqrt{3}} \|a\|_2 \leq \left(\frac{1}{|V|} \sum_{x \in V} |\langle a, x \rangle|^p \right)^{1/p} \leq \|a\|_2.$$

8. Short Khinchin inequality

Let $1 \leq p < \infty$. Recall that $\|\cdot\|_p$ denotes the standard ℓ_p norm in $\mathbb{R}^n$, and B_p^n its unit ball.

Let $X \in \mathbb{R}^n$ be a vector with independent centered random ± 1 coordinates, i.e. a random vertex of the discrete cube $\{-1, 1\}^n$. The classical Khinchin inequality, Theorem 3.4 asserts that for any $p \geq 1$ and for any vector $a \in \mathbb{R}^n$, $(\mathbb{E}|\langle a, X \rangle|^p)^{1/p}$ is equivalent to $\|a\|_2$ up to multiplicative constants depending on p . This equivalence can be obtained if one averages not over the whole discrete cube, but over some small part of it. The problem how small should this set be was around since mid-seventies. More precisely,

Let $p \geq 1$. Find constants α_p, β_p and a set $V \subset \{-1, 1\}^n$ of a small cardinality such that

$$\alpha_p \|a\|_2 \leq \left(\frac{1}{|V|} \sum_{x \in V} |\langle a, x \rangle|^p \right)^{1/p} \leq \beta_p \|a\|_2$$

for any $a \in \mathbb{R}^n$.

Deterministic constructions of sets V of reasonably small cardinality are unknown. Therefore, we shall construct the set V probabilistically. Namely, we choose $N = N(n, p)$ and consider N independent copies $X_1, \dots, X_N$ of the random vector X . If $N \ll 2^{n/2}$, in particular, if N is polynomial in n , all vectors $X_1, \dots, X_N$ are distinct with high probability. The problem thus is reduced to showing that with high probability, any vector $y \in \mathbb{R}^n$ satisfies

$$(8.1) \quad \alpha_p \|y\|_2 \leq \left(\frac{1}{N} \sum_{j=1}^N |\langle y, X_j \rangle|^p \right)^{1/p} \leq \beta_p \|y\|_2.$$

This problem can be recast in the language of random matrices. Let A be the $N \times n$ matrix with rows $X_1, \dots, X_N$. Then the inequality above means that A defines a

Figure 1: The short Khinchin construction problem on page 23 of the official arXiv PDF.

Thus the deterministic cardinality has the optimal $n^{p/2}$ order at every even $p > 2$. The theorem is partial relative to the sharp general problem: it gives $O(n^2)$ rather than the probabilistic $O(n)$ scale at $p = 1$, and rounds $p/2$ upward for non-even $p > 2$.

3 The BCH-type sign set

Fix $m \geq 1$. Let $Q = 2^r$ be the least power of two with $Q - 1 \geq n$, so $Q < 2(n + 1)$, and choose distinct nonzero $t_1, \dots, t_n \in \mathbb{F}_Q$. Write $\text{Tr} : \mathbb{F}_Q \rightarrow \mathbb{F}_2$ for the field trace. For a seed $u = (u_1, \dots, u_m) \in \mathbb{F}_Q^m$, set

$$\varepsilon_j(u) = (-1)^{\text{Tr}(\sum_{\ell=1}^m u_\ell t_j^{2^\ell - 1})}, \quad 1 \leq j \leq n. \quad (3)$$

Let $V_{n,m}$ be the set of distinct sign vectors $(\varepsilon_1(u), \dots, \varepsilon_n(u))$ as u varies. This is an explicit primitive-BCH orthogonal array in trace coordinates. Plainly

$$|V_{n,m}| \leq Q^m < [2(n + 1)]^m. \quad (4)$$

Lemma 2. *The uniform distribution on $V_{n,m}$ is $2m$ -wise independent.*

Proof. The bit map underlying (3) is linear over $\mathbb{F}_2$, so the uniform seed pushes forward to the uniform distribution on its image. It remains to prove that every set of at most $2m$ coordinate functionals is linearly independent.

Otherwise, for some nonempty $S \subset \{1, \dots, n\}$ with $s = |S| \leq 2m$, summing the corresponding bits would vanish for every seed. Nondegeneracy of the trace pairing then gives

$$\sum_{j \in S} t_j^{2^\ell - 1} = 0, \quad 1 \leq \ell \leq m. \quad (5)$$

In characteristic two, an even power sum is the square of the power sum at half its exponent. Iterating this fact and using (5) shows

$$\sum_{j \in S} t_j^h = 0, \quad 1 \leq h \leq 2m.$$

In particular these identities hold for $1 \leq h \leq s$. But the square matrix $(t_j^h)_{1 \leq h \leq s, j \in S}$ has determinant

$$\left(\prod_{j \in S} t_j \right) \prod_{i < j, i, j \in S} (t_j - t_i) \neq 0.$$

It cannot annihilate the all-ones vector. This contradiction proves the claimed independence. $\square$

4 Proof of Theorem 1

Let $p = 2m$. Expanding $(\sum_{j=1}^n a_j \varepsilon_j)^{2m}$ produces monomials involving at most $2m$ distinct signs. Lemma 2 says their expectations under the uniform measure on $V_{n,m}$ equal their expectations under independent Rademacher signs. Since the exponent is even, this proves the exact identity (2). The classical Khinchin inequality and (4) prove part 1.

For part 2, let $2m$ be the least even integer at least p . Pairwise independence gives, for $S_a(x) = \langle a, x \rangle$,

$$\|S_a\|_{L_2(V_{n,m})} = \|a\|_2.$$

Monotonicity of normalized finite-measure norms and part 1 give

$$\|a\|_2 = \|S_a\|_2 \leq \|S_a\|_p \leq \|S_a\|_{2m} \leq B_{2m} \|a\|_2.$$

Finally use $V_{n,2}$ when $1 \leq p \leq 2$. Its second and fourth moments match the full cube, so

$$\|S_a\|_2 = \|a\|_2, \quad \|S_a\|_4^4 = 3\|a\|_2^4 - 2 \sum_j a_j^4 \leq 3\|a\|_2^4.$$

Log-convexity between L_1, L_2, L_4 yields

$$\|S_a\|_1 \geq \frac{\|S_a\|_2^3}{\|S_a\|_4^2} \geq \frac{1}{\sqrt{3}} \|a\|_2.$$

Now $\|S_a\|_1 \leq \|S_a\|_p \leq \|S_a\|_2$, completing the proof.

5 Verification, novelty, and scope

The construction is algebraic and uses no numerical evidence. The key compression from $O(n^{2m})$ naive polynomial seeds to $O(n^m)$ is the odd-power BCH trace map; Lemma 2 includes the full Vandermonde verification. Passing from seeds to distinct sign vectors is legitimate because a linear map has equal-size fibers.

The run indexes, parsed corpus, exact phrase, and bounded searches through 11 August 2026 for short Khinchin constructions, BCH codes, orthogonal arrays, k -wise independent Rademachers, and explicit Euclidean subspaces were checked. Pass and Spektor [2] study Khintchine inequalities under k -wise independence, while coding theory standardly supplies the required arrays; no source directly presenting the synthesis above as an answer to Rudelson’s construction question was located. Novelty confidence is moderate-to-low because both ingredients are classical, although the stated consequence appears absent from the bounded search.

Human review should focus on the intended meaning of “reasonably small” in the survey. The even-exponent theorem achieves the source’s optimal order and is a sharp solved subcase. The all- p extension is polynomial but is not claimed sharp outside even exponents.

References

- [1] M. Rudelson, *Recent developments in non-asymptotic theory of random matrices*, arXiv:1301.2382 (2013).
- [2] B. Pass and S. Spektor, *On Khintchine type inequalities for k -wise independent Rademacher random variables*, Statistics and Probability Letters **132** (2018), 35–39; arXiv:1708.08775.